Lecture 22 (*April 12*)

Reading: Section 6.1
Viewing: Lecture 21

Just like in the last few classes, we're sticking with square matrices, so let A be an $n \times n$ matrix. The goal is to make the matrix "as simple as possible". What this means precisely might be the subject of some debate, but some of the simplest square matrices out there are diagonal ones:

$$\mathrm{diag}_{d_1,\dots,d_n} = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \tag{181}$$

Indeed, the linear transformation $\mathbb{R}^n \to \mathbb{R}^n$ that corresponds to $\mathrm{diag}_{d_1,\dots,d_n}$ simply scales along the unit vectors $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ of the coordinate axes (specifically, it scales by a factor of d_i along the i–th coordinate axis of $\mathbb{R}^n$). As we have seen in (159), we have $\det(\mathrm{diag}_{d_1,\dots,d_n}) = d_1 \dots d_n$, which reflects the fact that our linear transformation scales volumes by a factor equal to $d_1 \dots d_n$.

Definition 18. *A matrix is called* ***diagonalizable*** *if it is similar to a diagonal matrix, i.e.:*

$$A = V \cdot \mathrm{diag}_{d_1,\dots,d_n} \cdot V^{-1} \tag{182}$$

for some invertible $n \times n$ matrix V, and some numbers $d_1, \dots, d_n$.

The property (182) *means that the linear transformation $\mathbb{R}^n \to \mathbb{R}^n$ that corresponds to A scales by a factor of d_i in the direction of the vector $\boldsymbol{v}_i$, where $\boldsymbol{v}_i$ is the i–th column of V. Therefore, a diagonalizable matrix is diagonal in the basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ of $\mathbb{R}^n$.*

Remark. *Almost all $n \times n$ matrices are diagonalizable, and next week we will see what one can say about those matrices which are not.*

Suppose you knew that the matrix A is diagonalizable, but you didn't know the specific numbers d_i and the matrix V. You could reconstruct them from the notion of eigenvalues and eigenvectors:

Definition 19. *Given a $n \times n$ matrix A, a non-zero vector $\boldsymbol{v} \in \mathbb{R}^n$ is called an* ***eigenvector*** *if:*

$$\boxed{A\boldsymbol{v} = \lambda \boldsymbol{v}} \tag{183}$$

for some number λ called an ***eigenvalue****.*

Indeed, $\boldsymbol{v}$ is an eigenvector of A with corresponding eigenvalue λ if the linear transformation corresponding to A scales by a factor of λ in the direction of the vector $\boldsymbol{v}$. This story is very closely connected with (182): if the matrix A is equal to $V \mathrm{diag}_{d_1,\dots,d_n} V^{-1}$, then all the d_i's are eigenvalues of A and the corresponding eigenvectors are the columns of V.

So how many eigenvalues/eigenvectors are there? It's easy to see that if $\boldsymbol{v}$ is an eigenvector, then so is any multiple of $\boldsymbol{v}$. Moreover, if $\boldsymbol{v}$ and $\boldsymbol{w}$ are both eigenvectors corresponding to the same eigenvalue

λ, then $\boldsymbol{v}+\boldsymbol{w}$ is also an eigenvector corresponding to λ. Therefore, a better question would be "how many linearly independent eigenvectors are there"? Well, from the equation (183) we obtain:

$$(A-\lambda I)\boldsymbol{v}=0$$

Since $\boldsymbol{v}$ is supposed to be non-zero, this implies that the matrix $A-\lambda I$ should be singular (i.e. not invertible). Therefore, Fact 15 implies that:

$$\boxed{\det(A-\lambda I)=0} \tag{184}$$

for any eigenvalue λ of A. Conversely, any number λ for which (184) holds is an eigenvalue of A (try to think of an argument). This leads us to the following:

Definition 20. *The* ***characteristic polynomial*** *of A is:*

$$p(\lambda)=\det(A-\lambda I) \tag{185}$$

as a function of the variable λ. The roots (i.e. those numbers λ for which $p(\lambda)=0$) of the characteristic polynomial are the eigenvalues of A.

Although not very practical, you could in theory compute the characteristic polynomial by applying the big formula (167) for the determinant:

$$p(\lambda)=\det\begin{bmatrix} a_{11}-\lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22}-\lambda & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn}-\lambda \end{bmatrix}=\sum_{\{\sigma(1),\dots,\sigma(n)\}}^{\text{permutations}}(-1)^{\text{sgn }\sigma}a'_{1\sigma(1)}a'_{2\sigma(2)}\dots a'_{n\sigma(n)}$$

where $a'_{ij}=a_{ij}$ if $i\neq j$ and $a'_{ii}=a_{ii}-\lambda$. Since every summand in the right-hand side is a product of n numbers a'_{ij} for various i and j, no summand can contain more than n factors of λ. Moreover, the only summand which contains exactly n factors of λ is $a'_{11}a'_{22}\dots a'_{nn}=(a_{11}-\lambda)(a_{22}-\lambda)\dots(a_{nn}-\lambda)$. Therefore, the characteristic polynomial has degree n in λ and its top degree coefficient is $(-1)^n$:

$$p(\lambda)=(-1)^n\lambda^n+\alpha_{n-1}\lambda^{n-1}+\dots+\alpha_1\lambda+\alpha_0$$

The coefficients $\alpha_{n-1},\dots,\alpha_0$ depend on the entries of the matrix A. For example:

$$\alpha_{n-1}=(-1)^{n-1}\cdot\text{tr }A \tag{186}$$

where:

$$\boxed{\text{tr }A=a_{11}+a_{22}+\dots+a_{n-1,n-1}+a_{nn}} \tag{187}$$

is called the **trace** of the matrix A. Clearly:

$$\alpha_0=\det A \tag{188}$$

is just the determinant of A (this is just saying that the value of the characteristic polynomial at $\lambda=0$ is the determinant of A, which is obvious given the definition (185)). Because of Vieta's formulas, which express the sum/product of the roots of a polynomial in terms of its coefficients, we conclude the following:

Fact 17. *The sum of the eigenvalues of a matrix A is equal to the trace of A.*

The product of the eigenvalues of a matrix A is the determinant of A.

As an exercise, compute the eigenvalues of the matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix} \tag{189}$$

First let us form the characteristic polynomial:

$$p(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 1 \\ 5 & 2-\lambda \end{bmatrix} = (2-\lambda)(2-\lambda) - 5 \cdot 1 = \lambda^2 - 4\lambda - 1$$

The eigenvalues of A are the roots of this polynomial, which you can get by the quadratic formula:

$$\lambda = \frac{4 \pm \sqrt{4^2+4}}{2} = 2 \pm \sqrt{5} \qquad \Rightarrow \qquad \begin{cases} \lambda_1 = 2 + \sqrt{5} \\ \lambda_2 = 2 - \sqrt{5} \end{cases}$$

In general, because the eigenvalues of a 2×2 matrix are the solutions of a quadratic equation, you should expect to encounter square roots in their formulas. As we have seen in the example above, in the 2×2 case, one can deduce the formula for the characteristic polynomial from the facts that:

- it is quadratic and its leading coefficient is 1
- its linear term is minus the trace of the matrix (186)
- its constant term is the determinant of a matrix (188)

So the characteristic polynomial of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is:

$$p(\lambda) = \lambda^2 - \lambda(a+d) + (ad - bc) \tag{190}$$

For matrices of size 3×3 and higher, the characteristic polynomial is of order 3 and higher, so it is a bit more complicated to write it down and a lot more complicated to find its roots.

Remark. *In general, it is **NOT** true that the eigenvalues of $A+B$ are the sums of the eigenvalues of A and B, nor are the eigenvalues of AB the products of the eigenvalues of A and B.*

Now suppose that you know λ is an eigenvalue of a square matrix A; how do you find an eigenvector $\boldsymbol{v}$ such that $A\boldsymbol{v} = \lambda\boldsymbol{v}$? You already have the tools for this: just recast the eigenvector relation as $(A - \lambda I)\boldsymbol{v} = 0$, so all you need to do is to find $\boldsymbol{v} \neq 0$ in the nullspace of the <u>singular</u> matrix $A - \lambda I$.

For example, in the case of the matrix (189), let us look for an eigenvector $\boldsymbol{v}$ corresponding to the eigenvalue $\lambda = 2 + \sqrt{5}$. To this end, let us construct the matrix:

$$A - \lambda I = \begin{bmatrix} -\sqrt{5} & 1 \\ 5 & -\sqrt{5} \end{bmatrix}$$

and we will simply pick a vector $\boldsymbol{v} \in N(A - \lambda I)$. Remember that, in order to compute the nullspace of a matrix, the way to go is to put it in reduced row echelon form:

$$\begin{bmatrix} -\sqrt{5} & 1 \\ 5 & -\sqrt{5} \end{bmatrix} \rightsquigarrow \begin{bmatrix} -\sqrt{5} & 1 \\ 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -\frac{1}{\sqrt{5}} \\ 0 & 0 \end{bmatrix}$$

Therefore, an eigenvector $\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ can be obtained by solving the equation:

$$\begin{bmatrix} 1 & -\frac{1}{\sqrt{5}} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \qquad \Leftrightarrow \qquad v_1 - \frac{v_2}{\sqrt{5}} = 0$$

So one particular choice of eigenvector can be obtained by setting the free variable v_2 equal to 1, and solving for the pivot variable from the equation above: $v_1 = \frac{1}{\sqrt{5}}$. Therefore, an eigenvector of the matrix A corresponding to the eigenvalue $\lambda = 2 + \sqrt{5}$ is:

$$\boldsymbol{v} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ 1 \end{bmatrix}$$

Any multiple of $\boldsymbol{v}$ would be an equally good eigenvector.